Course milestone M13: Final Poster Lock

HONR 46400 · Evidence-Driven Research

Start here

Your main file is `lastname_m13.pdf`; the artifact listed at the end of this section travel beside it. Build it in this order:

1. **Your carry-forward record, at the very top.** Open my review email and your research action plan side by side. Answer every numbered request for this milestone (each has an ID like `IR-M13-01`) and every step your plan sets for this milestone. Next to each one, write **Done** and exactly where I can find it, or **Not adopted**, with your reason and what you chose instead. Several items can point to the same piece of work.
2. **Then your milestone work and files.** The book milestone below explains the research work. The complete course checklist, the files to hand in and the rubric are in this milestone's assignment on Brightspace: check them before you submit.
3. **The Expo work** in *What this course adds*, below.

How this counts. Your answer to the action plan is worth 10 points of the rubric, scored for you individually. Each numbered request left unanswered costs 5 points, up to 20. Only requests and plan steps I sent you at least 48 hours before the deadline count; anything later moves to the next milestone. A step that is also a numbered request is counted once, as a request. If I sent you no numbered requests, or no plan steps apply, say so in one line. Not sure what applies to you? Write to me before the deadline; that is exactly what I am here for.

Submit it on Brightspace, under **Assignments** then **M13**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 10: Your artifact, ready to publish or present** (revisit, version 3 - the locked print edition) of the course book, EDR|AI.

Beside your PDF, hand in:

- `lastname_m13_poster.pdf`: The print-ready poster at URC print spec. It goes straight to the press on a run shared with QM 47400, so it must ship as its own file at print dimensions and cannot be a section inside another PDF.

What this course adds

Your venue is fixed, and the lock is terminal

The Book Milestone lets you choose a venue. Here it is chosen for you: a printed poster at the Purdue Undergraduate Research Expo.

Write your venue contract against the URC print spec handed out in class, which sets dimensions, minimum font sizes at print scale, and the required identifying elements. The poster has to be legible at the size it will be printed, not only on your screen.

This lock is terminal. There is no revision window, because your file goes straight into a shared print run. The poster you submit is the poster strangers read at the Expo.

Send the poster through the URC channel as instructed in class as well. The course-platform timestamp is what governs the deadline. If you are in an approved group, the poster is one shared submission naming every member.

Milestone 10: Your artifact, ready to publish or present

Open the milestone workbook in Colab

This chapter closes Studio 10: Prepare to publish or present. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. A short venue contract and one artifact ready for publication or presentation that satisfies it - a paper or research note, a seminar or conference talk, a poster, or another venue's format - with a content map tracing every load-bearing claim and number, and a defense rehearsal record. This is not yet public release.

What you bring

You also carry forward the last milestone's artifact: **Milestone 9: Your bounded claims**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 35: Difficult Questions and Uncertainty: your five hardest fair questions with defend-or-concede answers, and your uncertainty-and-limitations statement in ULN form.

These pieces are conditional. Bring each one when its condition is yours; when it is not, the lesson's own drill or a plain skip is the right call, and the milestone does not require the piece.

- Lesson 32: Research Posters: if your venue is a poster, or your format carries a quantitative figure: the draft with labeled values, uncertainty shown as the design licenses (interval or caveat), and a headline verbed to your boundary.
- Lesson 33: Poster Criticism: if your venue is a poster: the gallery-walk audit of your draft, hits ranked by how far the error travels, fixed once, and the lock call.
- Lesson 34: Research Pitches, Talks, and Seminars: if you will speak, whether a pitch, a seminar, or a conference talk: your 30-second and 3-minute spines, built boundary-first, tested aloud on an outsider, and expanded to the venue's length.

Your format decides one more input. For a paper or research note, bring your Milestone 9 note; for a seminar or talk, the pitch spine; for a poster, the visual pieces above; for another format, the nearest row's input plus your venue contract. Difficult Questions is everyone's.

Choosing your format

The goal of this studio is an artifact ready for publication or presentation: a manuscript a venue could receive, a talk a room could hear, a poster a conference could hang. Start with a **venue contract**: your written record of what the audience will receive and what the venue allows. Name the audience, the medium, the length or space limit, the required elements, the accessibility needs, the submission rule, and the question channel. One line each is enough. Example: a five-minute recorded talk, six slides, captions required, no live questions.

Every format carries the same research core: the bounded claim, the evidence that earns it, uncertainty and limitations, traceable numbers and citations, and whatever disclosure or permission information the audience needs. The format only decides where those pieces live. Choose the row that fits your venue, work the lessons it names, and when you borrow a technique from another row, name the technique rather than claiming the whole lesson transfers.

If your venue asks for	build it with	then test and rehearse
a paper or research note	your Studio 9 note, developed to the venue contract: sections, length, references, and the figure rules for any figure	the gallery-walk audit from Poster Criticism, run on the manuscript; Difficult Questions in writing
a seminar or conference talk	Research Pitches' spine, expanded section by section to the venue's length; slides follow the poster lesson's figure rules	the gallery-walk audit from Poster Criticism, run on the deck; Difficult Questions, spoken
a short pitch	Research Pitches	Difficult Questions, spoken
a poster	Research Posters	Poster Criticism, then Difficult Questions
another format	the venue contract plus the nearest row, naming what transfers	Difficult Questions, in the venue's own mode

The poster lessons are this book's fully worked visual example: one surface, no narrator, every number exposed. The paper route runs on your Studio 9 note; the seminar route runs on the pitches lesson's spine. Difficult Questions is for everyone.

If your venue has no live questions, the rehearsal still happens: have a peer or an AI reviewer send you hard questions in writing, and record defend, concede, or do-not-know for each.

The practice

1. Write your venue contract using the guide above, then say what your claim must lose to fit that format.
2. Write a transformation memo from your Studio 9 note: what expands, what compresses, what moves, what becomes visual, and what becomes spoken.
3. Build a content map before you build the artifact: where the claim, the evidence, the boundary, the uncertainty, and the disclosure will each live.
4. Build the artifact to its venue contract, keeping every load-bearing number, figure, and citation traceable to the claim-evidence table.
5. Test it in its actual mode: scan the poster at gallery distance, time the talk aloud, advance the deck without narrating, or hand the manuscript to a cold reader.
6. Rehearse the defense against the three hardest questions you can construct, and record which you could not answer; those are the agenda for the studio that owns them.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** An adaptation that drops the limitations is not a shorter version, it is a different claim.
- **Evidence, provenance, and reproducibility.** Every figure that travels carries its verification with it.
- **AI activity, verification, and human decisions.** If AI edits the artifact or generates questions, record what changed and verify the result; every defend-or-concede decision remains yours.
- **Uncertainty, claim boundary, and revision history.** The shorter the format, the more explicit the boundary has to be.

Where this milestone sits

You arrived carrying Milestone 9: Your bounded claims. Milestone 11 lets strangers rerun what you have just defended, which is the one test rehearsal cannot give you. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

A question you cannot answer in rehearsal is evidence, and it sends you back to the studio that owns it.

The book's own checks

Your course rubric on Brightspace scores these checks inside its rows, so use this table to test your work before you hand it in. The points below are the book's, not your course score.

Each row scores **0**, **1**, or **2**. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking

#	Criterion	0	1	2
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: A short venue contract and one artifact ready for publication or presentation that satisfies it - a paper or research note, a seminar or conference talk, a poster, or another venue's format - with a content map tracing every load-bearing claim and number, and a defense rehearsal record. This is not yet public release	contract, adaptation, content map, or rehearsal absent, or a load-bearing claim exceeds its evidence	all pieces present, but one load-bearing number, boundary, accessibility path, or mode test is incomplete	the adaptation satisfies its declared contract, every load-bearing claim and number is traceable, uncertainty travels with the claim, and the rehearsal records defensible responses plus the unresolved work

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** `Runtime` → `Run all`, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

`File` → `Print` opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.